# MERGE: Multimodal Emotion-Driven Rhythmic Lyric Generation Engine

**Abhay Sharma, Ankit Kumar, Hritik Ranjan, Vikash Kumar**

## Abstract

Creative language generation has gained increasing attention in natural language processing, particularly for tasks such as poetry and song lyric generation. However, most existing systems focus either on structural constraints such as rhyme and meter or on semantic control mechanisms such as sentiment, topic, or style. The integration of multimodal emotional signals with rhythm-aware lyric generation remains relatively underexplored.

In this work, we propose **MERGE** (Multimodal Emotion-Driven Rhythmic Lyric Generation Engine), a framework for generating rhythm-aware song lyrics conditioned on emotional states inferred from facial expressions. The system first detects the user's emotional state using a facial emotion recognition model and then conditions a transformer-based language model on the predicted emotion to guide lyric generation. To maintain lyrical structure, rhythm-aware constrained decoding is applied to enforce syllable counts and rhyme patterns during generation.

We further introduce **REDS**, a composite evaluation metric designed for lyric generation that measures rhythmic consistency, emotional alignment, lexical diversity, and semantic coherence. MERGE provides a framework for exploring emotion-conditioned lyric generation while preserving structural properties of song lyrics.

## 1 Introduction

The motivation of this project is to develop an emotion-aware system capable of generating creative musical content based on a user's emotional state inferred from sensory inputs. Direct music synthesis conditioned on physiological or multimodal sensor data poses significant challenges due to limited dataset availability, the high dimensionality of audio generation, and the complex alignment between emotional intent and musical structure (Cao et al., 2022). To preserve the objective of emotion-driven creative generation while remaining within the methodological scope of natural language processing (NLP), this work reformulates the problem as emotion-conditioned lyric generation.

Facial emotion recognition provides a practical and widely studied modality for sensing emotional states (Goodfellow et al., 2013; Li et al., 2017). At the same time, song lyrics serve as a primary textual medium through which emotional expression is conveyed in music. These properties make lyrics a suitable domain for studying emotion-controlled and structurally constrained language generation. Prior research in NLP has explored emotion-conditioned text generation as well as structural constraints such as rhyme schemes and syllable patterns in poetry and lyric generation. However, these directions have largely been investigated independently.

A key challenge in emotion-conditioned lyric generation lies in the limited availability of large-scale lyric datasets with explicit emotion annotations. While curated datasets such as the Spotify Million Playlist Dataset (Devdope, 2023) provide approximately 500k lyrics annotated with emotional labels, their scale remains limited for training modern neural language models. To address this limitation, this work leverages large-scale lyric corpora obtained from the Genius lyric dataset (Lim and Benson, 2021), which provides substantially larger collections of song lyrics. Since explicit emotion annotations are not available for these datasets, we construct emotion labels using a scalable annotation pipeline that aggregates predictions from a GoEmotions-based classifier and maps fine-grained emotional categories into the six facial emotion recognition (FER) classes through normalization and weighted line-level aggregation. This procedure enables the creation of large-scale emotion-labeled lyric datasets suitable for training emotion-

conditioned language models.

While prior work has examined controlled text generation and rhythm-aware lyric generation separately, the integration of facial emotion signals as control inputs for rhythm-aware lyric generation remains relatively underexplored, particularly in English-language songwriting. Bridging this gap requires a framework that can incorporate visual emotional cues into the generation process while maintaining the structural characteristics that define song lyrics.

In this work, we propose **MERGE (Multimodal Emotion-Driven Rhythmic Lyric Generation Engine)**, a multimodal framework that connects facial emotion recognition with rhythm-aware lyric generation. MERGE uses facial emotion recognition models to infer emotional states from visual inputs and employs these signals to condition a transformer-based language model trained on large-scale lyric corpora. To ensure structural coherence, the generation process incorporates rhythm-aware decoding constraints based on syllable counts and rhyme schemes.

This work investigates two central research questions: (1) whether facial emotion signals can effectively serve as control inputs for lyric generation, and (2) whether rhythmic constraints such as syllable count and rhyme can be enforced during decoding in emotion-conditioned lyric generation without degrading emotional alignment or linguistic fluency. The contributions of this work are fourfold: (1) MERGE, a multimodal framework that connects facial emotion recognition with lyric generation, (2) a scalable lyric emotion annotation pipeline that maps fine-grained GoEmotions predictions to facial emotion recognition (FER) categories using normalization and weighted aggregation, enabling large-scale emotion labeling of lyric corpora, (3) a rhythm-aware constrained decoding strategy for maintaining lyrical structure during generation, and (4) REDS, a composite evaluation metric designed for assessing emotional alignment and structural quality in generated lyrics.

The source code and implementation details for this project are available at the following link.[1]

## 2 Related Work

**Facial Emotion Recognition.** Facial emotion recognition has achieved strong performance using deep convolutional neural networks, with architectures such as ResNet and MobileNet commonly evaluated on benchmark datasets including FER-2013 and RAF-DB (Goodfellow et al., 2013; Li et al., 2017). The FER-2013 dataset, introduced as part of a large-scale representation learning challenge, contains over 35,000 grayscale facial images annotated with basic emotion labels such as angry, disgust, fear, happy, sad, surprise, and neutral (Goodfellow et al., 2013). Due to its accessibility and scale, FER-2013 remains one of the most widely used benchmarks for training and evaluating facial emotion recognition systems. More recent research has explored transformer-based architectures and multimodal fusion networks for emotion recognition, improving the ability of models to capture contextual emotional signals from visual data (Zhang et al., 2023). These advances make facial emotion recognition sufficiently reliable to serve as an emotion sensing component in downstream generative systems.

**Controlled Text Generation.** In natural language processing, controlled text generation has been widely studied as a mechanism for steering language models toward desired attributes such as sentiment, topic, or style. Early approaches relied on conditional fine-tuning strategies and control tokens for large language models (Radford et al., 2019). More recent work has explored attribute-guided generation techniques that allow large language models to produce text aligned with specific semantic properties, including emotional tone and stylistic constraints (Hu et al., 2023). These approaches demonstrate that external control signals can effectively influence the behavior of generative language models.

**Lyric and Poetry Generation.** Research on poetry and lyric generation has extensively explored methods for enforcing structural constraints such as rhyme schemes, meter, and syllable counts. Early approaches relied on rule-based techniques and phonetic resources such as the CMU Pronouncing Dictionary to analyze syllable patterns and rhyming structures (Ghazvininejad et al., 2016; Gonçalo Oliveira, 2024). More recent neural approaches incorporate such constraints during generation to produce structurally coherent poetic text. However, most existing systems primarily focus on structural control and do not explicitly investigate how these rhythmic constraints interact with emotion-conditioned generation, particularly when emotional signals originate from exter-

[1] https://github.com/IIITH-2025-27/NLP_Project

nal modalities such as facial expressions. Neural approaches later introduced sequence-to-sequence and transformer-based architectures capable of generating structured poetic text. More recent lyric generation systems incorporate musical or prosodic signals to guide the generation process. For example, SongRewriter (Sun et al., 2023) employs a randomized multi-level masking strategy to enable both lyric generation and editing while incorporating keyword prompts and vowel modeling tasks to encourage flexible rhyme schemes. XAI-Lyricist (Liang et al., 2024) conditions lyric generation on musical prosody signals to improve singability while also providing interpretable explanations for rhythmic alignment.

**Collaborative and Multimodal Generation Systems.** Another emerging direction explores collaborative or multi-agent architectures for lyric generation. Liu et al. (Liu and Liu, 2024) propose a multi-agent system in which different agents handle specialized tasks such as rhyme suggestion, syllable control, and lyric–melody alignment. Similarly, Ram et al. (Ram et al., 2021) employ multitask transfer learning with the T5 transformer model to model lyrical constraints including beat alignment and target word endings. More broadly, recent multimodal generative models such as MusicLM (Huang et al., 2023) demonstrate the potential of combining semantic signals with generative models for music-related tasks, although these systems primarily focus on audio generation rather than lyric generation.

Despite these advances, most existing systems either focus on structural lyric generation or rely on musical signals such as melody or prosody as conditioning inputs. The integration of visual emotion signals—such as facial expressions—with rhythm-aware lyric generation remains relatively underexplored, particularly in English-language songwriting. MERGE addresses this gap by providing an end-to-end framework that connects facial emotion recognition with constrained lyric generation.

## 3 Methodology

This work proposes an end-to-end multimodal framework that connects facial emotion recognition with emotion-conditioned and rhythm-aware lyric generation. The system (as shown in Figure 1) first detects emotional states from facial inputs and then uses these signals to guide lyric generation while enforcing linguistic and rhythmic constraints such as rhyme patterns and syllable structure.

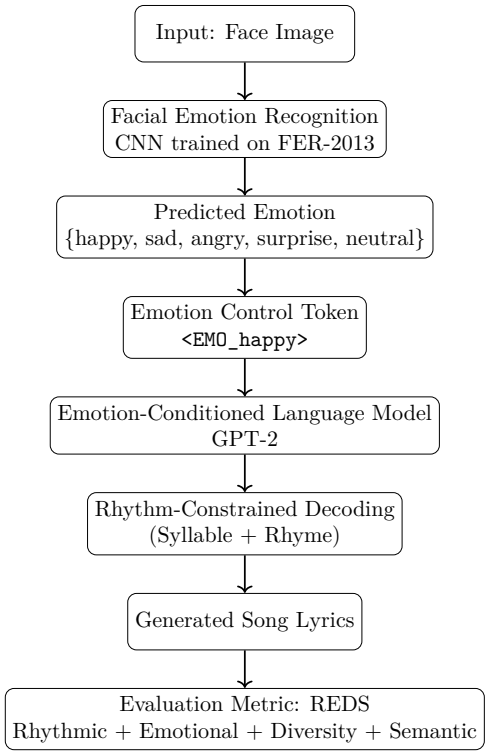


Figure 1: Overview of the MERGE framework for emotion-driven lyric generation.

### 3.1 Facial Emotion Conditioning

The MERGE framework uses facial expressions as the emotional input signal for lyric generation. Given a facial image, a facial emotion recognition model predicts the emotional state of the user (as shown in Figure 2), which is then used to condition the lyric generation process.

The facial emotion recognition model is trained on the FER-2013 dataset (Goodfellow et al., 2013), which contains 35,887 grayscale facial images labeled with basic emotion categories. Due to the severe class imbalance of the disgust category in FER-2013, the task is reformulated as a 6-class classification problem. The final emotion set used in this work is

$$\mathcal{E} = \text{angry}, \text{fear}, \text{happy}, \text{sad}, \text{surprise}, \text{neutral}.$$

Let $I$ denote the input facial image and let $f_\theta$ represent the facial emotion recognition model parameterized by $\theta$. The model produces a vector of logits

$$\mathbf{z} = f_\theta(I)$$

where $\mathbf{z} \in \mathbb{R}^6$ represents the unnormalized scores corresponding to the six emotion categories.

The probability distribution over emotions is computed using the softmax function

$$P(e_k \mid I) = \frac{\exp(z_k)}{\sum_{j=1}^{6} \exp(z_j)}$$

where $e_k \in \mathcal{E}$ denotes the $k$-th emotion category and $z_k$ represents the corresponding logit value.

The predicted facial emotion is obtained by selecting the class with the highest probability

$$e^* = \arg\max_{e_k \in \mathcal{E}} P(e_k \mid I)$$

The predicted emotion $e^*$ is then used as the conditioning signal for the lyric generation model. During inference, this emotion is converted into a control token (for example <sad>) which is prepended to the input sequence of the language model. This allows the model to generate lyrics that are aligned with the emotional state inferred from the user's facial expression.

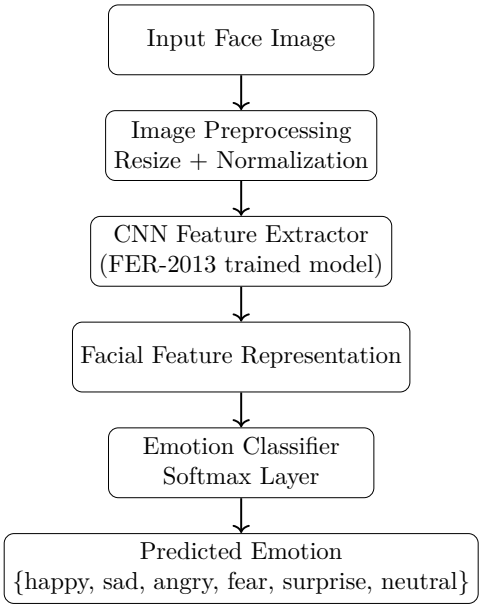


Figure 2: Pipeline of the facial emotion recognition module.

### 3.2 Dataset Construction and Emotion Mapping

Large-scale lyric datasets typically lack explicit emotion annotations, which makes it difficult to train emotion-conditioned language models. To address this limitation, we construct an emotion-labeled lyric dataset using an automatic annotation pipeline as shown in Figure 3.

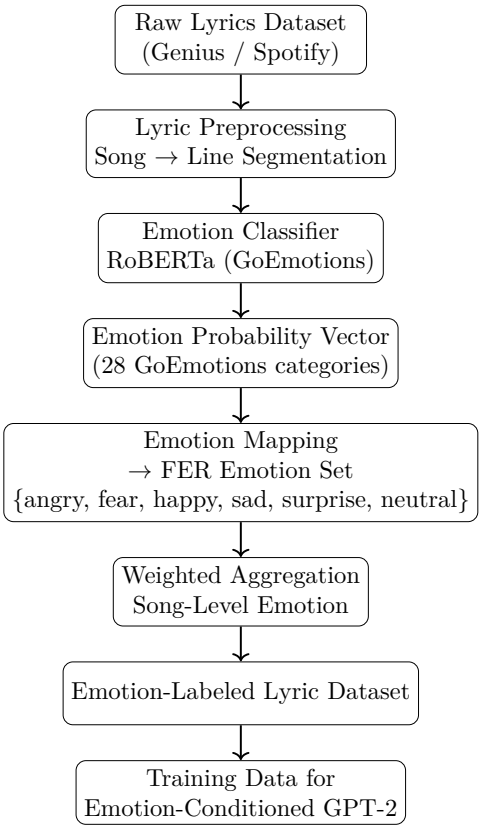


Figure 3: Pipeline for constructing the emotion-labeled lyric dataset.

Lyrics are collected from large public lyric corpora, specifically the Genius dataset (Lim and Benson, 2021), which contains lyrics in many languages out of which approx 3.2 million belongs to English language. Emotional signals are inferred using a pretrained emotion classifier based on the RoBERTa architecture(Lowe, 2023), which is fine-tuned on the GoEmotions dataset (Demszky et al., 2020). The GoEmotions label space contains 28 categories, consisting of 27 fine-grained emotion labels and one neutral category. For each input lyric line, the classifier outputs a probability distribution over these 28 labels, which serves as the basis for constructing the emotion representations used in the subsequent stages of the pipeline.

To ensure compatibility between textual emotions and facial emotion recognition outputs, the fine-grained emotion space is mapped into a smaller set of basic emotions, inspired by the cir-

cumplex model of affect and theories of basic emotions (Russell, 1980; Ekman, 1992). The emotion space is reduced to the following set

$$\mathcal{E} = \{\text{angry}, \text{fear}, \text{happy}, \text{sad}, \text{surprise}, \text{neutral}\}.$$

The disgust category is excluded for two reasons. First, explicit expressions of disgust rarely occur in song lyrics and are often linguistically expressed as sadness or disappointment. Second, the FER-2013 facial emotion dataset contains very few samples labeled as disgust, resulting in severe class imbalance during model training. To maintain a consistent emotion space across modalities, disgust-related signals are therefore merged into the sadness category. The created dataset is being hosted on kaggle(Kumar et al., 2025).

### 3.3 Lyric Dataset and Emotion Labeling

Each song is segmented into individual lyric lines. Let a song contain $n$ lyric lines, where the index of a line is denoted by

$$i \in \{1, 2, \ldots, n\}.$$

For each lyric line $i$, the GoEmotions classifier produces a probability vector over the 28 emotion labels:

$$\mathbf{p}_i = \left[p_i^{(1)}, p_i^{(2)}, \ldots, p_i^{(28)}\right] \in \mathbb{R}^{28},$$

where $p_i^{(e)}$ denotes the probability assigned to label $e$ and satisfies

$$0 \leq p_i^{(e)} \leq 1.$$

To obtain emotion representations compatible with facial emotion recognition categories, the fine-grained emotion labels are grouped into six emotion sets:

$$\begin{aligned}
A &= \{\text{anger}, \text{annoyance}, \text{disapproval}\},\\
F &= \{\text{fear}, \text{nervousness}\},\\
H &= \{\text{admiration}, \text{amusement}, \text{approval}, \text{caring},\\
&\quad \text{desire}, \text{excitement}, \text{gratitude}, \text{joy}, \text{optimism},\\
&\quad \text{love}, \text{pride}, \text{relief}\},\\
S &= \{\text{sadness}, \text{grief}, \text{disappointment}, \text{remorse},\\
&\quad \text{disgust}\},\\
U &= \{\text{surprise}, \text{realization}\},\\
N &= \{\text{neutral}, \text{curiosity}, \text{confusion}\}.
\end{aligned}$$

Using these groups, the emotion score for each category is computed by summing probabilities belonging to the same group. For example, the anger score for line $i$ is defined as

$$q_i^{\text{angry}} = \sum_{e \in A} p_i^{(e)}.$$

Applying the same procedure to all emotion groups produces a six-dimensional emotion vector for lyric line $i$:

$$\mathbf{q}_i = \left[q_i^{\text{angry}}, q_i^{\text{fear}}, q_i^{\text{happy}}, q_i^{\text{sad}}, q_i^{\text{surprise}}, q_i^{\text{neutral}}\right].$$

Because multiple probabilities are aggregated, the resulting vector is normalized to obtain a valid probability distribution:

$$\mathbf{q}_i = \frac{\mathbf{q}_i}{\sum_{k=1}^{6} q_i^{(k)}}.$$

Lyrics often contain narrative lines with little emotional content. To reduce their influence, each line is assigned an emotional weight defined as

$$w_i = 1 - q_i^{\text{neutral}},$$

where $w_i$ represents the emotional intensity of lyric line $i$.

The final emotion representation of a song is computed by aggregating the line-level emotion vectors using weighted averaging:

$$\mathbf{E}_{\text{song}} = \frac{\sum_{i=1}^{n} w_i \mathbf{q}_i}{\sum_{i=1}^{n} w_i}.$$

The dominant emotion of the song is defined as

$$e^* = \arg\max_k \mathbf{E}_{\text{song}}^{(k)},$$

where $\mathbf{E}_{\text{song}}^{(k)}$ denotes the score of emotion category $k$.

### 3.4 Emotion-Conditioned Language Model

Lyric generation is formulated as a controlled text generation task in which emotional context influences the semantic content of generated lyrics. We fine-tune the GPT-2 language model (Radford et al., 2019) on the emotion-labeled lyric dataset constructed in the previous subsection.

Emotion conditioning is implemented using explicit control tokens representing the target emotion category. During training, an emotion token (for example `<EMO_sad>`) is prepended to each lyric sequence so that the model can learn associations between emotional states and linguistic patterns.

Let

$$x = (x_1, x_2, \ldots, x_T)$$

denote a lyric token sequence of length $T$, and let $e$ represent the conditioning emotion label. The model learns the conditional probability

$$P(x \mid e) = \prod_{t=1}^{T} P(x_t \mid x_{<t}, e),$$

where $x_t$ denotes the token generated at step $t$ and $x_{<t}$ represents the previously generated tokens.

The model is trained using the standard cross-entropy objective for next-token prediction.

### 3.5 Rhythm-Constrained Decoding

To enforce rhythmic and rhyming structure during lyric generation, constrained decoding is applied. The CMU Pronouncing Dictionary provides phonetic representations that allow syllable counting and rhyme detection (CMU, 2014).

Each generated lyric line must satisfy predefined structural constraints such as a target syllable count and a specified rhyme scheme (for example AABB or ABAB). During decoding, candidate tokens that violate these constraints are filtered from the vocabulary.

When a rhyming token is required at the end of a line, the candidate token set is restricted to words whose phonetic representations match the desired rhyme pattern. Hard constraints enforce structural consistency within each line, while soft back-off strategies allow constraint relaxation or line regeneration if decoding becomes infeasible.

By integrating these constraints directly into the decoding process rather than applying them as post-processing steps, the system generates lyrics that maintain both rhythmic structure and semantic coherence.

## 4 Evaluation

Evaluating creative language generation is challenging because traditional metrics such as BLEU or ROUGE primarily measure lexical overlap and often fail to capture important qualities of lyrics such as emotional alignment, rhythmic structure, and semantic coherence. To address this limitation, we introduce a composite evaluation metric called **REDS**, designed specifically for emotion-conditioned lyric generation.

REDS evaluates generated lyrics across four complementary dimensions: rhythmic consistency, emotional alignment, lexical diversity, and semantic coherence.

### 4.1 REDS Evaluation Metric

Let a generated lyric consist of $n$ lines

$$L = l_1, l_2, \ldots, l_n.$$

The REDS score is computed as a weighted combination of four components:

$$\text{REDS} = \alpha R + \beta E + \gamma D + \delta S$$

subject to the normalization constraint

$$\alpha + \beta + \gamma + \delta = 1.$$

Here $R$, $E$, $D$, and $S$ denote rhythm consistency, emotional alignment, lexical diversity, and semantic coherence respectively, while $\alpha$, $\beta$, $\gamma$, and $\delta$ represent weighting coefficients.

#### 4.1.1 Rhythm and Rhyme Consistency ($R$)

Rhythmic structure is evaluated using syllable counts and rhyme patterns derived from phonetic representations in the CMU Pronouncing Dictionary (CMU, 2014). Let $s_i$ denote the number of syllables in lyric line $l_i$, and let $s^*$ denote the target syllable count for the desired rhythmic pattern.

The syllable consistency score is defined as

$$R_{\text{syllable}} = 1 - \frac{1}{n} \sum_{i=1}^{n} \frac{|s_i - s^*|}{s^*}.$$

Rhyme consistency is evaluated by checking whether each line satisfies the expected rhyme scheme. Let

$$r_i = \begin{cases} 1 & \text{if line } l_i \text{ satisfies the rhyme scheme,} \\ 0 & \text{otherwise.} \end{cases}$$

The final rhythm score combines both syllable and rhyme correctness:

$$R = \frac{1}{2} \left( R_{\text{syllable}} + \frac{1}{n} \sum_{i=1}^{n} r_i \right).$$

#### 4.1.2 Emotional Alignment ($E$)

Emotional alignment measures how well the generated lyrics reflect the intended conditioning emotion. An independent emotion classifier trained on the GoEmotions dataset (Demszky et al., 2020) is used to estimate the emotion distribution of the generated text.

Let

$$\mathbf{p} = [p_1, p_2, \ldots, p_K]$$

denote the predicted probability distribution over the emotion set

$$\mathcal{E} = \text{angry}, \text{fear}, \text{happy}, \text{sad}, \text{surprise}, \text{neutral}.$$

If the intended emotion is $e^*$, the emotional alignment score is defined as

$$E = p_{e^*}.$$

#### 4.1.3 Lexical Distinctiveness ($D$)

Lexical diversity is measured using distinct n-gram statistics. Let $U_k$ denote the number of unique $k$-grams in the generated lyrics and let $T_k$ denote the total number of $k$-grams.

The distinct-$k$ score is defined as

$$\text{Distinct}_k = \frac{U_k}{T_k}.$$

The lexical diversity score is computed as

$$D = \frac{\text{Distinct}_1 + \text{Distinct}_2}{2}.$$

#### 4.1.4 Semantic Coherence ($S$)

Semantic coherence measures the topical consistency between consecutive lyric lines. Let $\mathbf{v}_i$ denote the sentence embedding of line $l_i$ obtained using a pretrained sentence embedding model.

The semantic coherence score is defined as the average cosine similarity between consecutive lines:

$$S = \frac{1}{n-1} \sum_{i=1}^{n-1} \frac{\mathbf{v} * i \cdot \mathbf{v} * i + 1}{|\mathbf{v} * i||\mathbf{v} * i + 1|}.$$

#### 4.1.5 Normalization

Each component score is normalized to the interval

$$R, E, D, S \in [0, 1].$$

This ensures that the evaluation components operate on comparable scales and prevents any single component from dominating the overall metric.

Consequently, the REDS metric is bounded by

$$0 \leq \text{REDS} \leq 1.$$

#### 4.1.6 Weight Optimization

The weighting coefficients $\alpha$, $\beta$, $\gamma$, and $\delta$ determine the relative importance of each evaluation component. Rather than selecting these weights manually, they are estimated using correlation-based optimization with human evaluation scores.

Let $H$ denote the vector of human evaluation ratings for generated lyrics. The optimal weights are obtained by maximizing the correlation between REDS scores and human judgments:

$$(\alpha, \beta, \gamma, \delta) = \arg\max \rho(\text{REDS}, H),$$

where $\rho(\cdot)$ denotes Spearman's rank correlation coefficient.

This optimization ensures that the REDS metric aligns with human judgments of lyric quality. Since this work is currently in progress, the REDS metric is proposed as the evaluation framework for lyric generation. The final weighting parameters $\alpha$, $\beta$, $\gamma$, and $\delta$ will be estimated using correlation with human evaluation scores during the final stage of the project.

## 5 Progress So Far

The following components of the MERGE framework have been implemented and experimentally validated.

**Facial Emotion Recognition.** A convolutional emotion recognition model was trained on FER-2013 across four experimental phases, progressively refining the architecture and training strategy. Phase I evaluated ResNet18 via a 243-combination grid search, establishing that Adam with a low learning rate ($5\times10^{-5}$) and medium augmentation provided the best baseline (≈62% test accuracy). Phase II upgraded to ResNet34, improving to 62.6% while revealing persistent overfitting (8.5% train–test gap) due to the absence of explicit regularisation. Phase III introduced ResNet50 with 3-phase progressive unfreezing, label smoothing ($\varepsilon$=0.1), and dropout, reducing the gap to 4.5% and raising accuracy to 66.8%. Phase IV removed the severely imbalanced *disgust* class, reformulating the problem as 6-class classification over $\mathcal{E}$={*angry, fear, happy, sad, surprise, neutral*}, and retraining the top models on the cleaner split. The final best model (ResNet50, Phase IV) achieves **67.43%** test accuracy, a weighted F1

of 0.671. Per-class F1 scores reflect the relative difficulty of each emotion: Happy = 0.884, Surprise = 0.763, Neutral = 0.658, Angry = 0.618, Sad = 0.544, Fear = 0.448.

| Phase | Backbone | Test Acc | Train–Test Gap |
|---|---|---|---|
| I | ResNet18 | 0.620 | 8–12% |
| II | ResNet34 | 0.626 | 8.5% |
| III | ResNet50 (7-cls) | 0.668 | 4.5% |
| IV | ResNet50 (6-cls) | **0.674** | <2% |

Table 1: FER-2013 accuracy progression across experimental phases. The largest gain (+4.2 pp) came from Phase II→III through regularisation and progressive fine-tuning; removing the disgust class in Phase IV further improved macro F1 and model stability.

**Lyric Dataset Construction.** A large-scale emotion-annotated lyric dataset was constructed to support emotion-conditioned language model training. Raw lyrics were sourced from the Genius corpus, containing approximately **3.2 million English-language** songs; a cleaned version of this corpus is released on Kaggle.[2] Emotional signals were inferred automatically using the GoEmotions-to-FER annotation pipeline described in Section 3, which maps 28 fine-grained emotion labels to the 6-class FER set via grouped aggregation and weighted song-level normalisation. The resulting dataset — pairing each song with a dominant emotion label from $\mathcal{E}$ — is also publicly available on Kaggle.[3] This dataset constitutes the primary training resource for the emotion-conditioned GPT-2 model and will be used in all subsequent language model experiments.

**Baseline Language Model.** A baseline lyric generation model was implemented by fine-tuning GPT-2 on a subset of the Spotify lyric dataset with emotion control tokens (e.g., `<EMO_sad>`, `<EMO_happy>`) prepended to each training sequence. Preliminary experiments confirm that the model generates emotionally coherent lyric sequences across all six emotion categories, with observable differences in lexical choice and thematic content between conditions. For example, `<EMO_happy>` outputs tend toward imagery of light and celebration, while `<EMO_sad>` outputs favour loss, emptiness, and longing. The current system does not yet incorporate rhythm-aware constrained decoding; generated lines vary in syllable count and do not reliably adhere to rhyme schemes. Integration of the CMU Pronouncing Dictionary-based decoder is the primary remaining technical objective.

[2] https://www.kaggle.com/datasets/ankit477/genius-dataset-cleaned

[3] https://www.kaggle.com/datasets/ankitkumar477/lyrics-with-mapped-emotions

## 6 Tentative Timeline

The remaining work of the project as shown in Table 2 focuses on completing the lyric generation models, implementing the REDS evaluation metric, and conducting comparative experiments.

| Week | Planned Work |
|---|---|
| Week 1 | Train and fine-tune additional language models for emotion-conditioned lyric generation and implement rhythm-constrained decoding. |
| Week 2 | Generate lyric samples and conduct comparative experiments across different models and decoding strategies. |
| Week 3 | Conduct human evaluation to assess emotional alignment, rhythmic quality, and overall lyric quality. |
| Week 4 | Estimate REDS weighting parameters, perform final evaluation experiments, and prepare the final report and presentation. |

Table 2: Tentative timeline for completing the MERGE framework.

## References


Qian Cao, Xu Chen, Ruihua Song, Hao Jiang, Guang Yang, and Zhao Cao. 2022. Multi-modal experience inspired ai creation. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 1445–1454.

CMU. 2014. The CMU Pronouncing Dictionary. http://www.speech.cs.cmu.edu/cgi-bin/cmudict.

Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. 2020. Goemotions: A dataset of fine-grained emotions. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4040–4054.

Devdope. 2023. 900k spotify songs with lyrics, emotions & more. Kaggle.

Paul Ekman. 1992. An argument for basic emotions. *Cognition & emotion*, 6(3-4):169–200.

Marjan Ghazvininejad, Xing Shi, Yejin Choi, and Kevin Knight. 2016. Generating topical poetry. In *Proceedings of the 2016 conference on empirical methods in natural language processing*, pages 1183–1191.

Hugo Gonçalo Oliveira. 2024. Automatic generation of creative text in portuguese: an overview. *Language Resources and Evaluation*, 58(1):7–41.

Ian J Goodfellow, Dumitru Erhan, Pierre Luc Carrier, Aaron Courville, Mehdi Mirza, Ben Hamner, Will Cukierski, Yichuan Tang, David Thaler, Dong-Hyun Lee, and 1 others. 2013. Challenges in representation learning: A report on three machine learning contests. In *International conference on neural information processing*, pages 117–124. Springer.

Zhiting Hu, Yelong Shen, and 1 others. 2023. Attribute-guided text generation with large language models. *arXiv preprint arXiv:2306.10891*.

Cheng-Zhi Anna Huang, Andrea Agostinelli, Jordi Pons, and 1 others. 2023. Musiclm: Generating music from text. *arXiv preprint arXiv:2301.11325*.

Ankit Kumar, Vikash Kumar, Hritik Kumar, and Abhay Sharma. 2025. Lyrics with mapped emotions. Accessed on March 8, 2026.

Shan Li, Weihong Deng, and JunPing Du. 2017. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2852–2861.

Qihao Liang, Xichu Ma, Finale Doshi-Velez, Brian Lim, and Ye Wang. 2024. Xai-lyricist: Improving the singability of ai-generated lyrics with prosody explanations. In *IJCAI*, pages 7877–7885.

Derek Lim and Austin R Benson. 2021. Expertise and Dynamics within Crowdsourced Musical Knowledge Curation: A Case Study of the Genius Platform. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 15, pages 373–384.

Hong-Hsiang Liu and Yi-Wen Liu. 2024. Agent-driven large language models for mandarin lyric generation. In *2024 27th Conference of the Oriental COCOSDA International Committee for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA)*, pages 1–6. IEEE.

Sam Lowe. 2023. roberta-base-go_emotions.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, and 1 others. 2019. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9.

Naveen Ram, Tanay Gummadi, Rahul Bhethanabotla, Richard J Savery, and Gil Weinberg. 2021. Say what? collaborative pop lyric generation using multitask transfer learning. In *Proceedings of the 9th international conference on human-agent interaction*, pages 165–173.

James A Russell. 1980. A circumplex model of affect. *Journal of personality and social psychology*, 39(6):1161.

Yusen Sun, Liangyou Li, Qun Liu, and Dit-Yan Yeung. 2023. Songrewriter: A chinese song rewriting system with controllable content and rhyme scheme. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 12863–12880.

Yifan Zhang, Ming Chen, and 1 others. 2023. Multimodal emotion recognition with transformer-based fusion networks. *IEEE Transactions on Affective Computing*.